

BRADLEY DIRKS

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POSITIONS

Stony Brook University <i>Simons Instructor</i>	2023-2027 Stony Brook, NY
NSF Mathematical Sciences Postdoctoral Research Fellow <i>Stony Brook University</i>	2023-2027 Stony Brook, NY
Institute for Advanced Study <i>Member, School of Mathematics</i>	2024-2025 Princeton, NJ

EDUCATION

University of Michigan <i>PhD in Mathematics. Advisor: Mircea Mustață</i> Thesis: Using Mixed Hodge Modules to Study Singularities	2018-2023 Ann Arbor, MI
University of California <i>BS and MA in Mathematics</i>	2014-2018 Los Angeles, CA

RESEARCH INTERESTS

Algebraic Geometry, Singularities, $\mathcal{D}$ -modules, Hodge Theory

PUBLICATIONS

- Restrictions of Hodge modules using generalized V -filtrations**
with Qianyu Chen and Sebastián Olano. To appear in Adv. Math.
- Fourier transform and Radon transform for mixed Hodge modules**
To appear in Ann. Inst. Fourier
- Some applications of Microlocalization for LCI subvarieties**
Math. Ann. **392** (2025)
- Verdier specialization and restrictions of Hodge modules**
with Qianyu Chen and Morihiko Saito. Math. Z. **310** (2025)
- The minimal exponent of cones over smooth complete intersection projective varieties**
with Qianyu Chen and Mircea Mustață. Rev. Roumaine Math. Pures Appl. **70** (2025)
- The minimal exponent and k -rationality for locally complete intersections**
with Qianyu Chen and Mircea Mustață. J. Ec. polytech. Math. **11** (2024)
- V -filtrations and minimal exponents of locally complete intersection singularities**
with Qianyu Chen, Mircea Mustață and Sebastián Olano. J. Reine Angew. Math. **811** (2024)
- An introduction to V -filtrations**
with Qianyu Chen and Mircea Mustață. Handbook of Geometry and Topology of Singularities, Volume VII
- On V -filtration, Hodge filtration and Fourier Transform**
with Qianyu Chen. Selecta Math. (N.S.) **29** (2023)
- Minimal exponents of hyperplane sections: a conjecture of Teissier**
with Mircea Mustață. J. Eur. Math. Soc. **25** (2023)
- Upper bounds for roots of B-functions, following Kashiwara and Lichtin**
with Mircea Mustață. Publ. Res. Inst. Math. Sci. **58** (2022)
- The Hilbert series of Hodge ideals of hyperplane arrangements**
with Mircea Mustață. J. Singul. **20** (2020)

PREPRINTS

1. **Microlocal Bernstein-Sato polynomials on singular ambient varieties**
with Michael Perlman. Preprint.
2. **Filtrations on local cohomology, higher singularities, and the Injectivity Theorem**
with Qianyu Chen and Sebastián Olano. Preprint.
3. **F -nilpotence and Hodge filtrations beyond isolated singularities**
with Jakub Witaszek. Preprint.
4. **Simple characterizations and finite descent properties of some Hodge theoretic singularity invariants**
with Sebastián Olano and Debaditya Raychaudhury. Preprint.
5. **Multiplier modules, V -filtrations and Bernstein-Sato polynomials on singular ambient varieties**
Preprint.
6. **Hodge theory of secant varieties**
with Qianyu Chen, Sebastián Olano and Debaditya Raychaudhury. Preprint.
7. **Partial cohomologically complete intersections via Hodge theory**
with Qianyu Chen and Sebastián Olano. Preprint.
8. **Hirzebruch-Milnor classes of local complete intersections, minimal exponent, and applications to higher singularities**
with Laurențiu Maxim and Sebastián Olano. Preprint.
9. **A Hodge theoretic generalization of $\mathbb{Q}$ -homology manifolds**
with Sebastián Olano and Debaditya Raychaudhury. Preprint.

AWARDS

US Junior Oberwolfach Fellow	MFO, 2025
NSF Mathematical Sciences Postdoctoral Research Fellowship	National Science Foundation, 2023
Wirt and Mary Cornwell Prize	University of Michigan Mathematics Department, 2023
Rackham Predoctoral Fellowship	University of Michigan, Summer 2022-Winter 2023
Paul Daus Memorial Award	UCLA, 2018
Departmental Honors	UCLA Math Department, 2018
Summa Cum Laude	UCLA, 2018

INVITED TALKS

BIRS Workshop on Singularities, D-modules and Connections to Physics	Sep. 2025
<i>Hodge theory of the constant sheaf</i>	BIRS
Singularities in the Midwest, IX	April 2025
<i>Hodge theoretic properties of LCI singularities</i>	University of Wisconsin, Madison
AMS 2025 Spring Sectional Meeting	March 2025
<i>Singularities of Local Complete Intersections via Hodge Theory</i>	University of Kansas
JHU Algebraic Geometry Seminar	October 2024
<i>Minimal Exponent for LCI subvarieties</i>	Johns Hopkins University
Princeton University Algebraic Geometry Seminar	September 2024
<i>Minimal Exponent for LCI subvarieties</i>	Princeton University
Birational Geometry Seminar 2024	May 2024
<i>Recent results on minimal exponent for LCI subvarieties</i>	Online
CUNY Commutative Algebra and Algebraic Geometry Seminar	March 2024
<i>The minimal exponent for LCI subvarieties</i>	Graduate Center of CUNY
University of Utah Algebraic Geometry Seminar	February 2024
<i>The minimal exponent for LCI subvarieties</i>	University of Utah
Columbia University Algebraic Geometry Seminar	January 2024
<i>The minimal exponent for LCI subvarieties</i>	Columbia University
Winter School on New Applications of Mixed Hodge Modules	January 2024
<i>V-filtration and Hodge filtration in higher codimension</i>	Simons Center for Geometry and Physics

University of Toronto Algebraic Geometry Seminar <i>The minimal exponent for LCI subvarieties</i>	November 2023 University of Toronto
Harvard/MIT Algebraic Geometry Seminar <i>The minimal exponent for LCI subvarieties</i>	October 2023 Harvard University
Birational Geometry Seminar 2023 <i>Higher Du Bois and higher rational singularities for LCI varieties</i>	May 2023 Online
MAGGC <i>V-filtrations of $\mathcal{D}$-modules</i>	August 2022 UIC
Algebraic Geometry and Singularities Learning Workshop & Conference <i>Comparing V-filtration of an ideal with that of a general linear combination</i>	June 2022 UW Seattle
Algebraic Geometry Seminar <i>The Structure of Monodromic Mixed Hodge modules</i>	February 2022 Stony Brook University
DOCAS Seminar <i>Understanding the roots of b-functions</i>	August 2021 Online
UConn Algebra Seminar <i>Minimal Exponents and a Conjecture of Teissier</i>	April 2021 Online
Topology and Singularities Seminar in Madison <i>Minimal Exponents and a Conjecture of Teissier</i>	March 2021 Online

ORGANIZING

2026 Spring Eastern Sectional Meeting <i>Boston College</i>	March 2026 w/ Sebastián Olano, Debaditya Raychaudhury and Wanchun Shen
Learning Seminar on mixed Hodge modules <i>Princeton University</i>	Fall 2024 w/ Bhargav Bhatt, Sung Gi Park and Jakub Witaszek
AIM Workshop “Higher Du Bois and higher rational singularities” <i>American Institute of Mathematics. See website</i>	October 28 - November 1, 2024 w/ Radu Laza
Winter School on New Applications of Mixed Hodge Modules <i>Simons Center for Geometry and Physics. See website</i>	January 2024 w/ Christian Schnell
Student Algebraic Geometry seminar <i>University of Michigan</i>	Winter 2021-2023 w/ Devlin Mallory in Winter 2021, w/ Saket Shah 2022-2023
$\mathcal{D}$-modules and Representation Theory Minicourse <i>University of Michigan</i>	Summer 2022
Mixed Hodge Theory Minicourse <i>University of Michigan</i>	Summer 2020 w/ James Hotchkiss
Variations of Hodge Structure Reading Group <i>University of Michigan</i>	Winter 2020 w/ James Hotchkiss

TEACHING EXPERIENCE

Instructor <i>Stony Brook University</i>	Spring 2024 Stony Brook, NY
• MAT310: Linear Algebra	
Graduate Student Instructor <i>University of Michigan</i>	October 2018-Winter 2022 Ann Arbor, MI
• Math 105 (Precalculus), Math 115 (Calculus I), Math 116 (Calculus II)	
Docent <i>LA Math Circle (now ORMC)</i>	2016 - 2018 Los Angeles, CA